

Highlights

Coordinate Feature Engineering for Enhanced Real-Time Hand Gesture Recognition Using Machine Learning

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- Recognizes gestures using coordinate data, not images, for efficient processing.
- Achieves good accuracy with classic ML models like KNN and Decision Trees.
- Provides a lightweight framework with potential for real-time applications.

Coordinate Feature Engineering for Enhanced Real-Time Hand Gesture Recognition Using Machine Learning

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ABSTRACT

Hand Gesture Recognition (HGR) is a core part of human-computer interaction (HCI) in a touchless manner. Traditionally, image-based HGR approaches are computationally expensive and vulnerable to lighting, background and device variations. Hence this work proposes a coordinate-based HGR framework that recognizes hand gesture by leveraging hand landmark coordinates extracted from hand images rather than directly using the image data. After data pre-processing and feature engineering, the extracted coordinate features were used to train and evaluate a number of ML classifiers like Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Decision Tree (DT), Naive Bayes (NB) and XGBoost. Based on the coordinate features the developed framework could perform gesture classification, activity recognition and person-activity classification. It is observed from the experiments that the coordinate-based approach can recognize the hand gestures precisely and efficiently while largely decreasing the computation complexity. It outperforms the existing state-of-art methods by providing higher accuracy and greater efficiency. Since the algorithm is light weighted and device independent it could be used for real-time applications on portable, embedded, edge computing and IOT devices. The further research work involves using large data sets, dealing with user and environment variations and incorporating Deep learning approaches for increased accuracy, robustness and scalability.

1. Introduction

Hand gesture recognition is a new and effective way to allow humans to interact with computers naturally and contactlessly. It enables the user to interact with digital systems through natural hand movements. Gesture-based interfaces offer an intuitive, touchless method of interaction, instead of the traditional input devices including keyboards, mice and touchscreens. The development of efficient and reliable gesture recognition systems is motivated by the growing demand for smart environments, assistive technologies, virtual applications and contactless systems.

Most of existing gestures recognition methods are based on image processing and vision techniques. These methods often demand high computational resources and are susceptible to factors such as lighting conditions, background complexity and camera variations. Processing full image frames adds unnecessary info and makes it hard to use in real time. Figure 1. Hand gesture activities in this study and their corresponding gesture classes. The different hand gestures are mapped to a command: Up, Down, Play, Pause and Stop. The figure illustrates how various hand shapes and finger postures can be translated into meaningful gesture commands for classification. Hand Gesture Recognition is

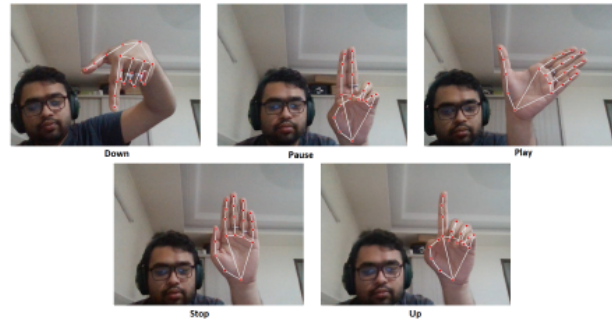


Figure 1: Visual representation of activities using hand gestures

very useful in our life. This allows people to communicate with machines without touching them. This is for people who cannot use their hands. It is also useful in houses where people can control things by gestures. Hand Gesture Recognition makes the machines simple to use for people.

The aim of this study is to design an efficient hand gesture recognition system based on the spatial coordinates of hand landmarks. The proposed method extracts geometric features which accurately represent hand posture and movement from three-dimensional hand landmark coordinates (X, Y, Z). This enables a reliable classification of gestures and low computational complexity. The coordinate based representation enables low-complexity and accurate gesture recognition which makes the system applicable for real-time control applications.

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1.1. Motivation and Objective

The project aims at providing a more natural touchless and intuitive way of interacting with computers and digital systems. As technology becomes more and more present in people's lives, traditional input devices such as keyboards, mice and touch screens may not always be the best option for enhancing interactivity for all. This is where factors like convenience, accessibility and hygiene come into play. The gesture-based and touchless interaction will certainly increase the usability of technology for a large part of the population, in particular where hands-free control is required, in assistive technologies for people with physical limitations, in virtual environments and in other smart systems. This will enhance the interaction between human and technology. However, from the technical point of view, working with raw visual input could be computationally heavy and rather inefficient in real-time and resource-poor systems. So this project is trying to solve this difficulty using hand landmark coordinate data instead of fully process images. The computational complexity is reduced and the required information for accurate gesture recognition is kept by using spatial coordinates of salient hand points on hand.

1.2. Challenges and Contribution

The key difficulty in this research was finding a way to turn a series of coordinates for hand landmarks into useful features without having complete images or videos of those hand landmarks. The challenge to achieve a high degree of accuracy is due to the fact that there are a limited number of possible values for the various coordinate positions. Also, because of various hand geometrical shapes, the way people gesture with their hands, how a person moves and so forth, can also negatively impact the recognition performance of the systems being used in this research when identifying a person's hand gestures. This research also aims to assure that the system being developed will operate properly in real-time and have low computational resources with reliable output.

To achieve this goal, this research proposes a new way to recognize hand gestures based only on coordinates (no image processing) and delivers a significant reduction in computation time. An organized data pre-processing and feature extraction method is developed for transforming raw coordinates into useful features to enable hand gesture recognition to occur in a timely manner. Various machine learning models will be trained using different training/testing ratio combinations to evaluate the consistency of the machine learning method for recognizing hand gestures. Furthermore, this research demonstrates that with minimal input data, we can achieve accurate hand gesture recognition; therefore, the proposed system can be used in real-time applications and in environments with limited computational resources.

The Rest of the Paper is organized as follows: In the Section 2 literature review is discussed. Section 3 describes the methodology; Section 4 Discuss the experimental

Setup; Section 5 Describes the Result and Discussion, Section 6 Finally Concludes the Paper.

2. Literature Review

In this section, we review the contributions of various researchers who have advanced hand gesture detection from coordinate data up to 2022. We study it in three aspects section 2.1 is Deep Learning and Computer Vision Based Approach, section 2.2 is IOT Sensor Based Approach and section 2.3 is Machine Learning Based Approach.

2.1. Deep Learning and Computer Vision Based Approach

Several researchers have leveraged deep learning and computer vision techniques to achieve high accuracy in hand gesture detection and recognition. Mohamed et al. (2021) Mohamed, Mustafa and Jomhari (2021) presented a comprehensive review of vision-based hand gesture recognition systems published between 2014 and 2020, emphasizing data acquisition methods, operating environments, and gesture representation techniques. Their study reported that signer-dependent systems achieved an accuracy of 88.8%, while signer-independent systems reached 78.2%, highlighting the need for improved continuous gesture recognition. Mujahid et al. (2021) Mujahid, Awan, Yasin, Mohammed, Damaševičius, Maskeliūnas and Abdulkareem (2021) proposed a real-time hand gesture identification model using YOLOv3 with DarkNet-53 CNN architecture. The model demonstrated superior performance compared to SSD and VGG16 in terms of accuracy, precision, recall, and F1-score, making it suitable for real-time applications, including assistive technologies for people with disabilities. Bhushan et al. (2022) Bhushan, Alshehri, Keshta, Chakraverti, Rajpurohit and Abugabah (2022) compared multiple classifiers for gesture recognition and observed that while Random Forest performed well, CNN-based models consistently outperformed others in image-based gesture recognition tasks, emphasizing their effectiveness. Benitez-Garcia et al. (2021) Benitez-Garcia, Prudente-Tixtco, Castro-Madrid, Toscano-Medina, Olivares-Mercado, Sanchez-Perez and Villalba (2021) introduced a real-time hand gesture recognition framework without optical flow computation by combining RGB frames and manual segmentation masks using FASSD-Net. Their approach improved accuracy, and the Temporal Shift Module (TSM) outperformed the Temporal Segment Network (TSN) while maintaining real-time performance. Sahoo et al. proposed a score-level fusion and fine-tuning strategy for pre-trained CNN models to address sparse tagging in static datasets, achieving excellent recognition performance in real-time American Sign Language (ASL) systems. Zhang et al. Zhang, Tian and Zhou (2018) proposed a deep learning framework combining convolutional neural networks and LSTM for dynamic hand gesture recognition in human-robot interaction, effectively learning both short-term and long-term temporal features. Devineau et al. Devineau, Moutarde, Xi and Yang (2018) presented a CNN-based deep learning model for 3D hand motion recognition, achieving

accuracies of 91.28% and 84.35% on the DHG dataset, outperforming existing methods. Abavisani et al. (2019) introduced a multi-modal 3D-CNN framework with SSA loss and focal regularization, achieving state-of-the-art performance on dynamic hand gesture datasets. Oyedotun et al. (2017) applied deep learning architectures such as CNNs and stacked denoising autoencoders to recognize 24 hand gestures, outperforming earlier approaches with improved accuracy and reduced error rates. Li et al. (2019) proposed a CNN-based gesture detection model that mitigates background and illumination challenges using unsupervised learning, reduced parameter complexity, and SVM-based classification to enhance robustness. Ye et al. (2018) and Ye, Tian, Huenerfauth and Liu (2018) focus on recognizing American Sign Language from continuous video streams, a task inherently suited for deep learning architectures like Convolutional and Recurrent Neural Networks. Similarly, the comprehensive survey by Rastgoo et al. (2021) extensively reviews deep learning advancements in the field, confirming its centrality in contemporary research. Work like that of Koller et al. (2015) and Koller, Forster and Ney (2015) on large-vocabulary continuous recognition also aligns with this data-driven, statistical modeling approach often implemented with deep learning frameworks.

The main advantages of this method are high recognition accuracy, real time performance and automatic feature extraction. It can effectively recognise static and dynamic hand gestures, which can improve human-computer interaction and applications in assistive technology. The main disadvantages are high computational requirements, dependence on large labelled datasets, and sensitivity to lighting variations and complex backgrounds. It can also degrade its performance for out-of-sample users, and training such models is usually time-consuming and computationally expensive.

2.2. IOT Sensor Based Approach

IoT and sensor-based approaches have also played a significant role in hand gesture detection, particularly in non-vision-based environments. Tian et al. (2018) introduced WiCatch, a device-free hand gesture recognition system that utilizes channel state information (CSI). By employing data fusion techniques and SVM classification, the system achieved an accuracy exceeding 96% and demonstrated effectiveness in recognizing two-hand gestures. Zhang et al. (2020) proposed Latern, a radar-based real-time hand gesture recognition system that integrates a recurrent 3D convolutional neural network with a connectionist temporal classification (CTC) algorithm. The system achieved a high accuracy of 96%, outperforming existing radar-based gesture recognition solutions and enabling robust real-time performance. Jaramillo et al. (2020) provided a comprehensive review of real-time hand gesture recognition using electromyography (EMG) signals and machine learning techniques. Their

analysis covered model architectures, data preprocessing, feature extraction, classification methods, and evaluation metrics, identifying key trends and research gaps in EMG-based gesture recognition systems. Sombandith, Walairacht and Walairacht (2017), Chai, Wang, Yin and Chen (2015).

IoT and sensor-based hand gesture recognition systems provides high accuracy with fast Execution Speed and work well in the Dark environment. However, these systems often require special sensors or devices, which can increase cost. Their performance may also be affected by signal interference, and some systems need extra training or calibration for different users.

2.3. Machine Learning Based Approach

Traditional machine learning approaches continue to contribute significantly to hand gesture recognition, particularly when combined with effective feature extraction techniques. Haria et al. (2017) developed a robust gesture recognition system for natural human-computer interaction capable of tracking both static and dynamic hand gestures. The system enabled real-world applications such as application control and presentation navigation with minimal hardware requirements. Lu et al. (2017) proposed a novel feature vector combined with a Hidden Conditional Neural Field (HCNF) classifier for recognizing dynamic hand gestures using the Leap Motion controller. Their approach achieved recognition accuracies of 89.5% and 95.0%, offering an efficient solution for dynamic gesture identification tasks. Additionally, Bhushan et al. (2022) demonstrated that ensemble-based machine learning models such as Random Forest provide competitive performance in gesture recognition, although deep learning models generally yield superior accuracy in complex visual datasets. Fagiani, Principi, Squartini and Piazza (2015), Lim, Tan and Tan (2016).

The traditional machine learning-based hand gesture recognition system is easy, inexpensive, and computationally inexpensive compared to deep learning approaches. It has high accuracy if used along with efficient feature extraction approaches, and it can be used in real-time for presentation control and human computer interaction. The disadvantage with this approach is that its efficiency is based on the feature extracted from the images and will work efficiently for only complex gestures.

3. Methodology

Feature selection is an Important Part of the hand gesture recognition system being proposed. Illustrated in Figure 3, features are picked from hand landmarks coordinates instead of full images or video frames. The hand gestures are represented as the x and y coordinates of main hand points, which are the major elements in describing the hand's shape and position. To reduce the amount of unnecessary data and improve the model's performance, only the most significant hand landmarks were used. By using coordinate-based features, the system reduces computational costs and

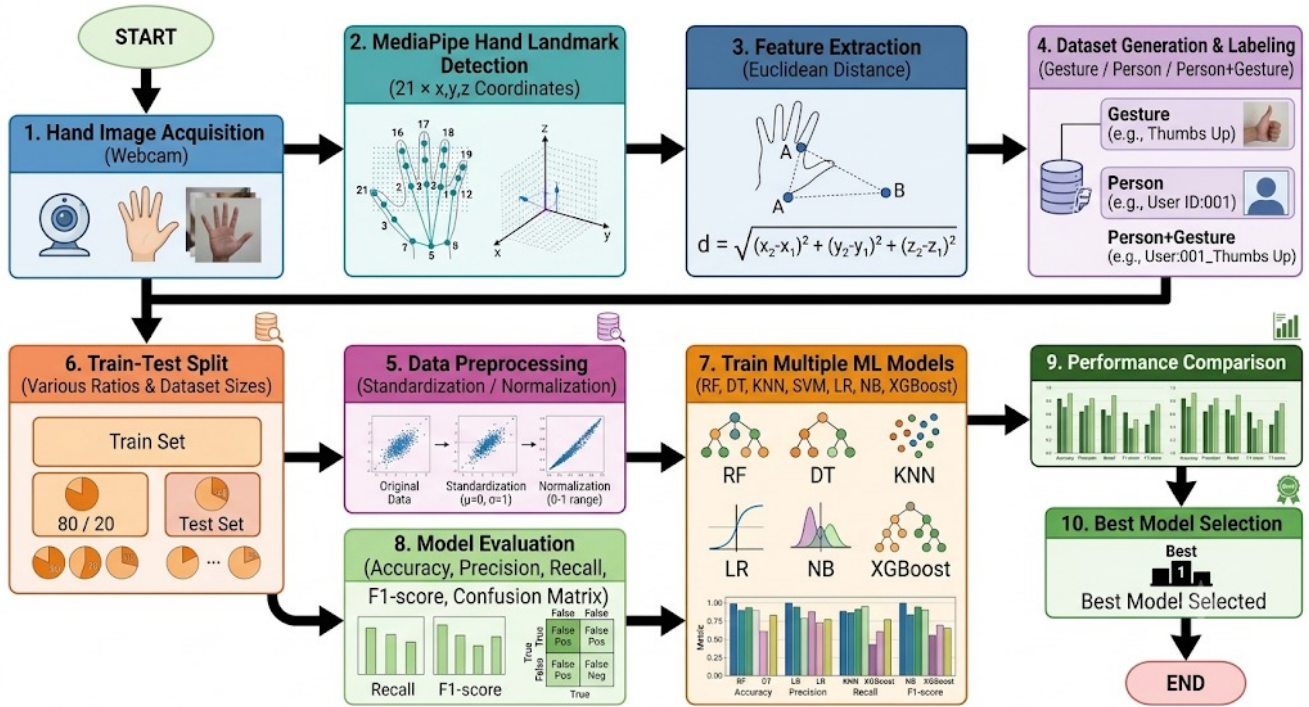


Figure 2: Overall workflow of the proposed person identification framework.

becomes both faster and more efficient. Such a technique also limits noise and offers better consistency when differences in gestures occur. The features chosen are able to represent major hand movements and at the same time keep the feature set simple and light. By not using complicated image features, the system is not only more straightforward to train, but also better tailored for the devices with the limited processing power. This feature selection tactic allows for highly accurate gesture recognition without sacrificing real-time performance.

3.1. Data Collection

The initial stage in the methodology includes collecting the data of hand gestures. With the use of a webcam, the gestures of more than 50 subjects have been recorded under controlled circumstances. Every gesture has been captured as a set of 21 hand landmark points, which contain the coordinates (x, y, z) of each key point of the hand joint. Such information provides spatial accuracy in recording hand movements and can be used as an input to feature extraction. Recorded data has been stored in JSON format for efficient management.

3.2. Data Preparation

After gathering the data, we pre-processed the gathered gesture data for machine learning analysis. Euclidean distances between pairs of landmarks were calculated using the JSON files containing coordinates of landmarks to compute useful features. These distances reveal significant features such as finger lengths, joint spacing and overall hand shape. For quality reasons the dataset was cleaned

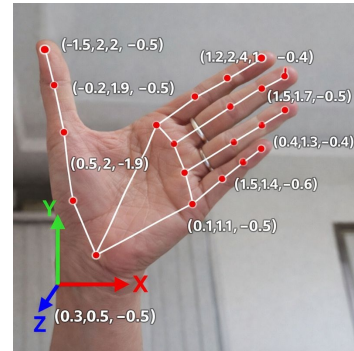


Figure 3: Three-Dimensional Hand Landmark Localization

by removing duplicates, correcting inconsistencies and handling missing values by careful checking and recollection when needed. We then structured the cleaned features into a dataset with structured labels creating a reliable and normalised dataset ready for training, validation and evaluation of gesture recognition models.

1. Removal of Invalid Data Elements

During data preparation, an evaluation was carried out to identify invalid data elements and consequently to eliminate the faulty data elements. Invalid data could be due to missing values, corrupted samples and repeated elements which would lead to inaccurate results for the model performance. When data values were missing, the same samples were again collected from the same individuals where it was possible and the unreliable samples were removed.

2. Preparation of Dataset and Data Normalization

The process involved in the preparation of the hand gesture dataset followed a two-stage approach where data were first collected and then properly cleaned and validated. In the data collection stage, hand gestures were recorded from over 50 subjects using a webcam and these data were recorded in JSON format where each sample had 21 hand landmarks each consisting of three coordinate points (x, y, z) . Feature extraction involved calculation of the Euclidean distance between pairs of the 21 hand landmarks. For two landmark points $p_1 = \{x_1, y_1, z_1\}$ and $p_2 = \{x_2, y_2, z_2\}$, their Euclidean distance was computed using the equation below.

$$d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Where x, y, z correspond to horizontal, vertical and depth axes respectively. This feature extraction process enables representation of geometric information of the hand like finger length, distance between joints, etc. In the data cleaning process, invalid or corrupted data samples were identified and re-collected from the same individuals, invalid data were removed and the dataset was prepared in a properly organized manner which is suitable for use in training and testing machine learning models.

Normalization refers to the scaling of the features to a fixed range, usually between 0 and 1. This approach is beneficial when there is a variance in the scales of the features and in the speed of convergence.

Normalization:
$$x_{\text{nor}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Standardization transforms the features so that they have an average value of 0 and a standard deviation of 1. This is used in the case of machine learning algorithms that are sensitive to normal distributions.

Standardization:
$$x_{\text{stand}} = \frac{x - \mu}{\sigma}$$

- x : original feature value,
- $x_{\min}$: minimum feature value,
- $x_{\max}$: maximum feature value,
- x_{nor} : normalized feature value,
- μ : average feature value,
- σ : standard deviation of the features,
- x_{stand} : standardized feature value.

After we cleaned the dataset we made it suitable for machine learning analysis. The features we extracted like the distance between hand landmarks were adjusted to have the scale. This way we did not give importance to features with bigger numbers. Each gesture sample had the label and the dataset was ready, for training, validation and testing. This helped us to effectively train and evaluate the gesture recognition model using the dataset.

3.3. Feature Selection

Characterizing features is a crucial component of the hand gesture recognition system that we propose. In the picture 3, firstly the features are derived from hand landmark coordinates rather than using complete images or video frames. The result of each hand gesture is a set of x and y coordinates of the key hand points that form the hand shape and its position. Most relevant hand landmarks only are chosen for leaving out the unnecessary information and enhancing the performance of the model. Selecting coordinate-based features leads to the reduction of computational cost and That means the system becomes faster and more efficient. Besides this, it also helps to lower the amount of noise and to make different gestures more consistent to each other. The features that are being chosen hereby are able to capture major hand movements without sacrificing the simplicity and lightness of the feature set. Since image features of a complicated nature are not used, the system is easier to train and more appropriate for those devices with lower processing power. Because of this, this feature selection technique allows for precise gesture recognition without compromising real-time performance.

3.4. Selection of Machine Learning Approaches

In this work, Recognition of human hand gestures using several machine learning models was carried out. Our data has numbers and categories such as up down, play, pause and stop so we need to use classification algorithm. In this research, we have used Logistic Regression, KNN, Decision Tree, Random Forest, XGBoost Naive Bayes and SVM as machine learning models. The reason we choose these models is that they use different learning methods and are also good at numerical feature classification problems. Then, the most appropriate model for hand gesture recognition can be determined by comparing their performances. Logistic Regression and Naive Bayes models are easy. Work quickly. KNN is useful as it will let us compare gestures. Decision Tree models are tree models. Tree models can handle patterns. So tree models are often more accurate than simple models. We added SVM because it is a classifier. We are using machine learning models to find the performance of the machine learning models and also to find the best and efficient way of detecting the gestures using machine learning models. We'll employ machine learning models to figure out which one is the best.

3.5. Model Validation

Model validation gives us a way to assess the quality of the machine learning algorithms for their performance on new and unseen data. For this purpose the entire dataset has been divided into training and testing sets, so that models can be trained on one portion of the data and tested on another. The effect of amount of training data on the results has been studied by varying the amount of training data in different training-testing ratios like 50-50, 60-40, 70-30 and 80-20. In addition, the authors have used the k-fold cross-validation method to ensure that the results are unbiased and reliable.

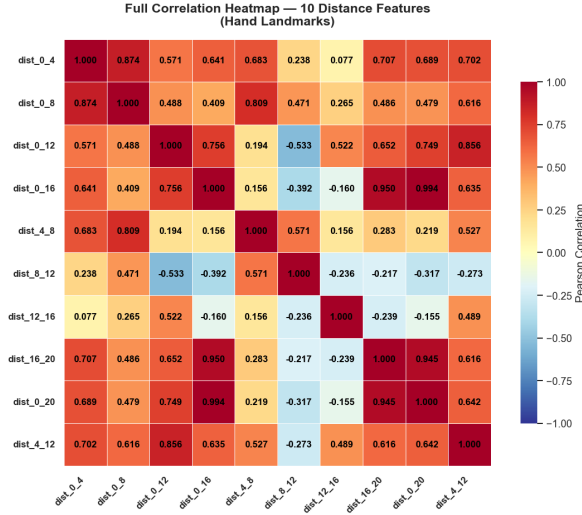


Figure 4: Visualization of Pairwise Feature Correlations

In this procedure, the dataset is split into a number of equal-sized parts, and the model is trained and tested multiple times, each time using a different part as the validation set. This will reduce bias and prevent overfitting. The efficacy of the machine learning algorithms has been checked by using accuracy, precision, recall and F1- score metrics. Accuracy, however, tells us how often the classifier is correct, but precision and recall take an extra step to measure the quality of the classifier. The good thing about the F1-score is that it is a kind of compromise between precision and recall. All these validation works allow us to compare the various classifiers and confirm the selected approach is robust, precise and practical for the recognition of real-time hand gestures.

$$\text{Accuracy} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i}$$

$$\text{Precision} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FP_i}$$

$$\text{Recall} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FN_i}$$

$$\text{F1} = \frac{1}{C} \sum_{i=1}^C \frac{2 \times TP_i}{2 \times TP_i + FP_i + FN_i}$$

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

4. Experimental Setup

To evaluate the gesture based classification framework in a systematic manner, two different experimental conditions

have been developed. One of these conditions tests the effect of varying the ratio of training to testing samples on the classification performance of gestures alone and person aware based on the different feature representation methods. The other condition studies the impact that varying the number of samples in the dataset will have on classification performance for gesture only and person aware representations. The entire dataset will provide complete evidence of the robustness, scalability and data efficiency of the proposed gesture based classification model through multiple experimental setups.

4.1. Prediction Activity Identification

After the addition of the person-specific variables, all the models showed a considerable improvement in performance. These extra elements helped to differentiate motions performed by various users and improve recognition accuracy. The Random Forest based models outperformed all the classifiers and showed robustness across varied dataset settings. Person-aware features also improved simpler classifiers significantly. Overall, the results show that the addition of gesture features with person-specific information can increase the classification accuracy, improve the resilience of the model and deliver more reliable predictions in real-world hand gesture recognition applications.

1. Variation of Training Ratio with Fixed Data Size:

The overall size of the dataset is maintained constant in this experiment. The training and testing ratios are different as 60:40, 70:30, 80:20 and 90:10. The aim is to see how the gesture recognition model works given amounts of training data. When the model is fed training data it learns from more instances thus it typically gets better at recognising gestures and interpreting what they mean. The experiment also seeks to determine the point where adding training data does not make a substantial difference in how well the model does. This means that we can find the

Table 1
Performance Metrics Across Different Train–Test Ratios

Accuracy							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.786	0.892	0.932	0.933	0.940	0.971	0.963
60–40	0.784	0.879	0.929	0.924	0.938	0.968	0.963
70–30	0.776	0.878	0.928	0.931	0.945	0.969	0.967
80–20	0.775	0.889	0.937	0.940	0.947	0.970	0.976
Precision							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.819	0.895	0.933	0.935	0.941	0.971	0.964
60–40	0.816	0.882	0.930	0.926	0.941	0.968	0.963
70–30	0.810	0.882	0.929	0.934	0.947	0.969	0.968
80–20	0.817	0.891	0.938	0.942	0.949	0.971	0.976
Recall							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.786	0.892	0.932	0.933	0.940	0.971	0.963
60–40	0.784	0.879	0.929	0.924	0.938	0.968	0.963
70–30	0.776	0.878	0.928	0.931	0.945	0.969	0.967
80–20	0.775	0.889	0.937	0.940	0.947	0.970	0.976
F1-Score							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.782	0.891	0.932	0.933	0.940	0.971	0.963
60–40	0.780	0.879	0.929	0.924	0.938	0.968	0.963
70–30	0.771	0.878	0.928	0.932	0.945	0.969	0.967
80–20	0.773	0.889	0.936	0.941	0.947	0.970	0.976

ratio of training data to use. The size of the data set is unchanged. The differences in model performance are due to various training/testing ratios, not because we have more data. When adjusted the ratios of training and testing as from 60:40, to 90:10 then the performance of the gesture recognition model varies. This tells us how the gesture recognition model works with training data ratios.

2. Variation of Data Size with Fixed Training Ratio:

In this experiment the size of the overall data set is increased but the ratio between training and testing is unchanged (e.g., 70:30 or 80:20). The aim is to analyse the impact of augmenting the quantity of gesture data on the performance of the gesture recognition model. The experiment was conducted using different dataset sizes (i.e., 100, 500, 1000, and 2000 samples) while keeping the same training–testing ratio. This approach allows to study data efficiency, performance saturation and model robustness with various dataset sizes. It also highlights the point where additional data produces little gain in recognition accuracy. The training ratio is fixed such that the performance differences are only due to increasing dataset size. The results give useful insights into the data needs and scalability of user-independent hand gesture detection systems.

4.2. Person Identification

This experiment tests the capacity of machine learning models to identify a person by features of hand gestures. The goal is to correctly identify the user, whose gesture data is being used, unlike gesture recognition, where the goal is to

classify the performed motion. The studies study the effects of varying the training ratio and dataset size on the accuracy and reliability of human identification.

1. Variation of Training Ratio with Fixed Data Size:

In this experiment we keep the size of the dataset the same and change the training and testing ratios. We try out ratios like 60:40, 70:30, 80:20 and 90:10 for the training and testing data. The main goal of this experiment is to see how the person identification works when we have amounts of training data. When we have training data the models get better at learning the patterns that are specific to each person, which means the person identification gets more accurate and it can identify people it has not seen before. We also want to find out at what point adding training data does not make a big difference in how well the person identification works. Since we are not changing the size of the dataset the changes we see in how well it works are only because we changed the ratio of training and testing data, for person identification.

2. Variation of Data Size with Fixed Training Ratio:

In this experiment, the training to testing ratio is. The amount of the data collection increases gradually. We look at dataset sizes to study the effect of more training samples on person identification performance. The data helps the models learn more about the particular manner each person writes and this improves the performance of the person identification. This means that the individual identification is more accurate, and it works well even when situations are not perfect. We also want to know when additional data do not

Table 2

Performance Comparison for Different Dataset Sizes(No Of Sample) with Fixed 80–20 Split (Experiment 2)

Accuracy							
Data Size	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
1000	0.825	0.857	0.845	0.875	0.976	0.975	0.965
1500	0.754	0.822	0.865	0.881	0.922	0.935	0.950
2000	0.757	0.836	0.889	0.898	0.934	0.947	0.945
3000	0.824	0.853	0.922	0.928	0.954	0.968	0.958
4000	0.835	0.920	0.952	0.950	0.956	0.977	0.971
5226	0.775	0.889	0.937	0.941	0.947	0.970	0.976
Precision							
1000	0.872	0.891	0.876	0.903	0.979	0.978	0.969
1500	0.788	0.838	0.881	0.895	0.927	0.940	0.956
2000	0.823	0.842	0.895	0.903	0.940	0.951	0.948
3000	0.842	0.870	0.929	0.934	0.956	0.970	0.961
4000	0.854	0.921	0.953	0.951	0.957	0.977	0.971
5226	0.817	0.891	0.938	0.943	0.949	0.971	0.976
Recall							
1000	0.825	0.857	0.845	0.875	0.976	0.975	0.965
1500	0.754	0.822	0.865	0.881	0.922	0.935	0.950
2000	0.757	0.836	0.889	0.898	0.934	0.947	0.945
3000	0.824	0.853	0.922	0.928	0.954	0.968	0.958
4000	0.835	0.920	0.952	0.950	0.956	0.977	0.971
5226	0.775	0.889	0.937	0.941	0.947	0.970	0.976
F1-Score							
1000	0.811	0.849	0.838	0.868	0.976	0.974	0.964
1500	0.750	0.821	0.863	0.882	0.922	0.933	0.950
2000	0.754	0.834	0.888	0.899	0.934	0.947	0.944
3000	0.821	0.854	0.922	0.928	0.954	0.968	0.958
4000	0.835	0.920	0.952	0.950	0.956	0.977	0.970
5226	0.773	0.889	0.936	0.942	0.947	0.971	0.976

improve individual identification very lot. The findings from this experiment reveal a lot about the data we need to develop precise and reliable person identification systems. This is something we need to know in order to develop systems for human identification. The person identification systems require data to perform well and via the experiment we learn the importance of quality data for person identification.

4.3. Prediction Activity with Person Identification

Both gesture features and person identification information are used in training and testing in this experiment. The dataset also contains an additional identifier for the person doing the gesture, as well as hand gesture features. It allows the model to learn user-specific gesture patterns and hand movement variations. The aim of this experiment is to investigate if the addition of person related information improves the classification performance compared to gesture alone recognition. This setting enables to evaluate the impact of person-aware features on model accuracy, stability and personalisation, by comparing results to the previous experiment, particularly in scenarios where gestures may differ across individuals.

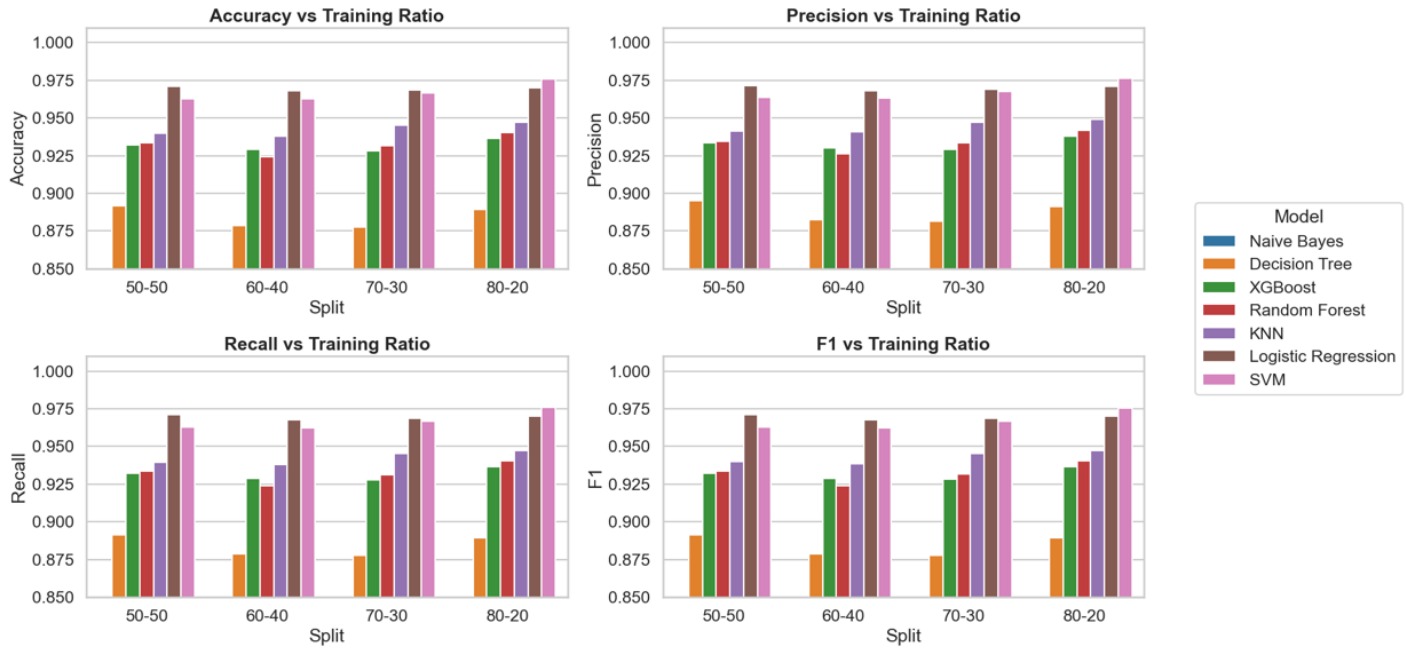
1. Variation of Training Ratio with Fixed Data Size:

In this experimental setting, the total size of the entire dataset is fixed and the proportion of samples used

for training and testing is systematically varied. Ratios like 50:50, 60:40, 70:30 and 80:20 for training-testing are considered. This setting is different from gesture-only classification because the model can leverage both gesture features and person identification information to learn personalised gesture patterns under various levels of training data availability. The purpose of this experiment is to analyse the effect of increasing the amount of training data on the classification accuracy, stability and generalisation with person-aware features. This setup isolates the effect of varying the training ratio by keeping the dataset size fixed and allows us to investigate the model's sensitivity to data distribution, and its tendency to overfit or improve personalisation with more training data.

2. Variation of Data Size with Fixed Training Ratio:

In this experimental setting, the training-testing ratio is fixed (e.g., 80:20) and the overall dataset size is increased incrementally. The size of the dataset is varied, but the same proportional split between training and testing samples is maintained. Gesture-related features and person identification features are covered in the whole experiment. This configuration tests the impact of access to more data that is aware of persons on the model performance. It allows analysis of data



Experiment 1: Model Performance Comparison Across Training Ratios

Figure 5: Model Performance Comparison Across Training Ratios

Table 3

Performance Metrics Across Different Train-Test Ratios

Accuracy							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.906927	0.828779	0.952009	0.985151	0.958285	0.981860	0.816762
60–40	0.934481	0.849354	0.960497	0.986514	0.964610	0.981923	0.847728
70–30	0.948342	0.872577	0.962883	0.986480	0.971301	0.983291	0.867474
80–20	0.955258	0.879541	0.963862	0.987763	0.973614	0.982218	0.881262
Precision							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.973039	0.853011	0.960231	0.987075	0.964123	0.984289	0.787287
60–40	0.976469	0.870112	0.968248	0.988973	0.970477	0.985344	0.825726
70–30	0.978075	0.890927	0.971161	0.989726	0.975648	0.987573	0.838801
80–20	0.976912	0.897434	0.971172	0.990880	0.978149	0.984681	0.853597
Recall							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.906927	0.828779	0.952009	0.985151	0.958285	0.981860	0.816762
60–40	0.934481	0.849354	0.960497	0.986514	0.964610	0.981923	0.847728
70–30	0.948342	0.872577	0.962883	0.986480	0.971301	0.983291	0.867474
80–20	0.955258	0.879541	0.963862	0.987763	0.973614	0.982218	0.881262
F1-Score							
Train Ratio	Naive Bayes	Decision Tree	XGBoost	Random Forest	KNN	Logistic Regression	SVM
50–50	0.926135	0.821949	0.950783	0.984488	0.954621	0.981079	0.785408
60–40	0.945807	0.842674	0.959441	0.985914	0.961804	0.981336	0.820960
70–30	0.954877	0.864273	0.961375	0.985971	0.968862	0.982676	0.840337
80–20	0.957678	0.869711	0.960715	0.986822	0.971160	0.980285	0.855633

scalability, learning efficiency and performance saturation in dependence on dataset size. Additionally, this experiment evaluates if the model can leverage the person-specific information to gain more effectively from larger amounts of data than gesture-only classification. The results from this setting provide insight

into the data requirements for developing robust and personalised hand gesture recognition systems that scale well with an increase in user participation.

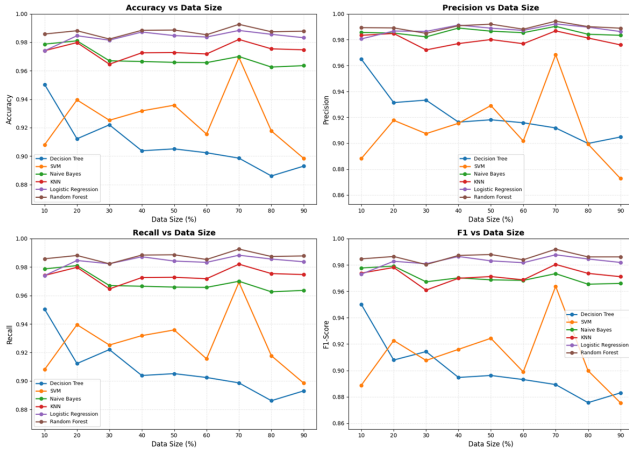


Figure 6: Variation of Data Size With 80- 20 Ratio (Gesture and Person Detection)

4.4. Variation of Training Ratio with Fixed Data Size

In this experimental condition, the overall size of the dataset remains constant while the ratio of training and testing samples is varied. This setting is used to investigate the impact of increasing the quantity of training data on classification accuracy and stability without altering the underlying data distribution. Four training – testing ratios (50 – 50, 60 – 40, 70 – 30, 80 – 20) are considered to represent the progressively increasing learning capacity. This condition was used in Experiment 1 and Experiment 3. Experiment 1 considers only gesture-based features and test model sensitivity to training ratio variation in a gesture-only setting. In Experiment 3, the person identification features are fused with the gesture information with the same data size and training ratios. This enables a direct comparison between gesture-only and person-aware classification under the same data conditions. The results for this condition provide insight into the robustness of the models, overfitting behaviour, and the effectiveness of feature augmentation as data availability varies during training.

4.5. Variation of Data Size with Fixed Training Ratio

In this experimental setting, the training-testing ratio is kept at 80-20 and the size of the dataset is changed to see how the volume of data affects the model's performance. We investigate the impact of adding more training samples on scalability, learning efficiency, and performance saturation. A constant training ratio makes performance differences mainly due to the size of the dataset and not due to sampling imbalance. This condition is implemented in Experiment 2 and in Experiment 4. In Experiment 2 we study the gesture-only classification on dataset sizes ranging from 1000 to 5226 samples, shedding light on the advantage of having more gesture data with different models. In Experiment 4 we extend this analysis with person and gesture features and increasing dataset sizes (from 10 percent to 90 percent of total data) and increasing number of subjects and gesture

classes. This condition allows us to assess model robustness for large-scale settings and to identify the performance saturation point. The results illustrate the data efficiency of the different models and the benefits of person-aware feature integration in achieving high classification accuracy with a limited amount of data.

5. Results and Discussions

The results of the experiment show that the gesture-based classification framework works under different conditions. We tested it with amounts of training data, dataset sizes and feature settings. The accuracy of classification got better as used in training data and larger datasets. This was true in all the experiments. After a point of improvement the accuracy became smaller. The benefits of adding data were limited but the framework still worked effectively with smaller datasets.

5.1. Prediction Activity Identification

The classification results of gestures show that the coordinate-based features are very useful to differentiate the hand activities performed. This is reflected in the high accuracy of the models such as Logistic Regression and SVM and the ability of the extracted features to sufficiently represent the underlying gesture pattern. Performance was very similar across different training and testing data splits, indicating consistent learning in spite of different configurations. The classification accuracy increased with the increasing size of the dataset, especially for the ensemble based models. Simple models achieve very good performance quickly. Probabilistic models tend to achieve lower accuracy as their features are not entirely independent of each other.

1. Variation of Training Ratio with Fixed Data Size:

The models exhibited stable generalisation across different performance was not significantly affected by changing the ratio (e.g. 60:40 to 90:10), with the only slight drop when the test set size was smaller. This demonstrates that the feature representation generalises well and the models do not overfit much even with larger relative training portions. Logistic Regression and SVM were the least sensitive to the variations of the ratio and were suitable for gesture recognition with limited dependence on the split configuration.

2. Variation of Data Size with Fixed Training Ratio:

As the total size of the dataset increased with a fixed training ratio (e.g. 80:20) clear and measurable improvements of classification accuracy were obtained. This was especially the case for ensemble methods like Random Forest and Gradient Boosting, which gained a lot from having more training examples. Ensemble and kernel-based approaches on the other hand kept improving, showing their ability to capture nuanced gesture patterns. Simpler models saturated sooner. The results highlight the importance of the dataset size for higher performance in gesture-only recognition tasks.

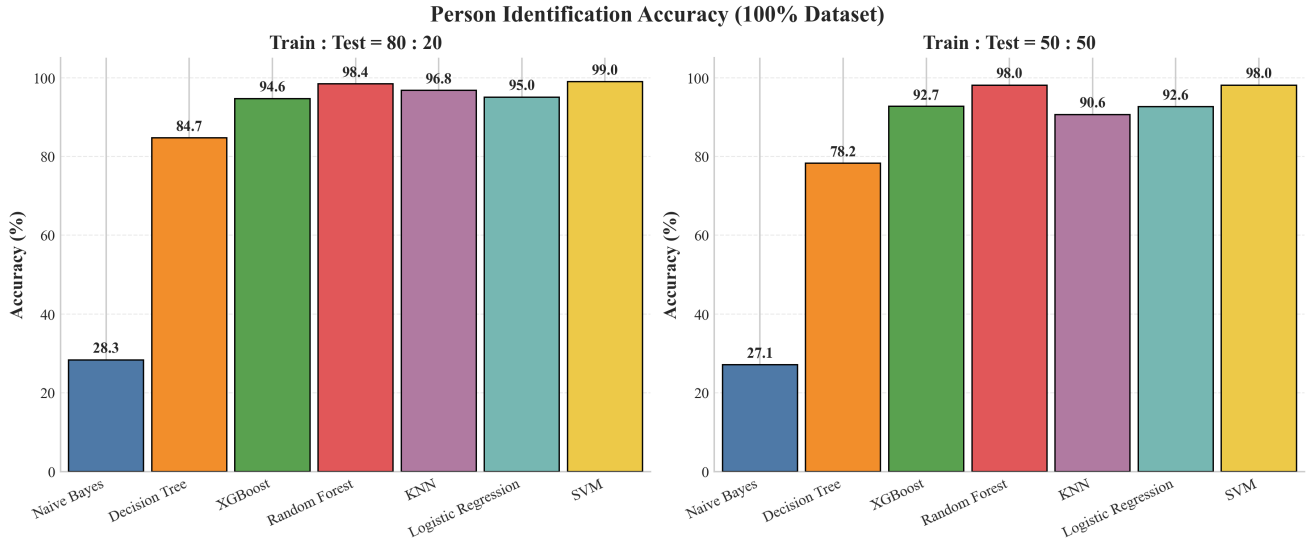


Figure 7: Person identification accuracy comparison across different dataset sizes for fixed train–test ratios.

5.2. Person Identification

The person identification results demonstrate that the proposed framework effectively distinguishes different individuals using the extracted hand landmark features. The coordinate-based representation captures unique spatial characteristics of each person’s hand, enabling reliable identification across multiple machine learning classifiers. The models exhibited consistent performance under different experimental settings, indicating that the extracted features provide strong discriminative information for identifying individuals.

1. Variation of Training Ratio with Fixed Data Size:

The person identification models were evaluated using different training-testing ratios with the dataset size fixed. The results show that the classification accuracy is stable with all the training ratios, which demonstrates good generalisation capability. Adding more training data only resulted in marginal gains as models were able to learn person specific characteristics with relatively less training samples. The most stable results were obtained using Logistic Regression, Support Vector Machine (SVM) and Random Forest, indicating that the proposed feature representation is robust and not prone to significant overfitting under different split configurations.

2. Variation of Data Size with Fixed Training Ratio:

The effect of dataset size was studied by performing experiments with increasing dataset sizes, keeping a fixed training–testing ratio of 80:20. The results show that the identification accuracy is increasing with the increasing number of training samples. Ensemble Classifiers (Random Forest and Gradient Boosting) gained the most from larger datasets, as they could learn more representative decision boundaries for individual users. There was also a steady improvement for kernel-based methods such as SVM.

The results show that the larger the dataset, the more robust and reliable the person identification. Thus, the proposed framework is appropriate for practical applications of biometric authentication and user identification.

5.3. Prediction Activity with Person Identification

When person-related features were incorporated, classification performance improved significantly across all models. The additional identity information helped reduce ambiguity between similar gestures performed by different users. Ensemble-based classifiers, particularly Random Forest, achieved the highest accuracy and showed strong robustness against data variations. Linear models also benefited from person-aware features and maintained reliable performance even with smaller training sets. These results confirm that combining gesture and person information enhances model discrimination capability and leads to more accurate and dependable predictions in real-world scenarios.

1. Variation of Training Ratio with Fixed Data Size:

Models leveraging person-aware features exhibited enhanced stability across varying training–testing splits. Unlike the gesture-only setting, performance showed even less fluctuation when the training ratio was altered, indicating that identity information provides a strong regularization effect. Random Forest and other ensemble methods maintained nearly consistent accuracy, even when the training set was reduced relative to the test set. This suggests that person-specific patterns are highly informative and can be learned efficiently, reducing the model’s dependence on large training portions for reliable generalization.

2. Variation of Data Size with Fixed Training Ratio:

Increasing the total dataset size with person-inclusive

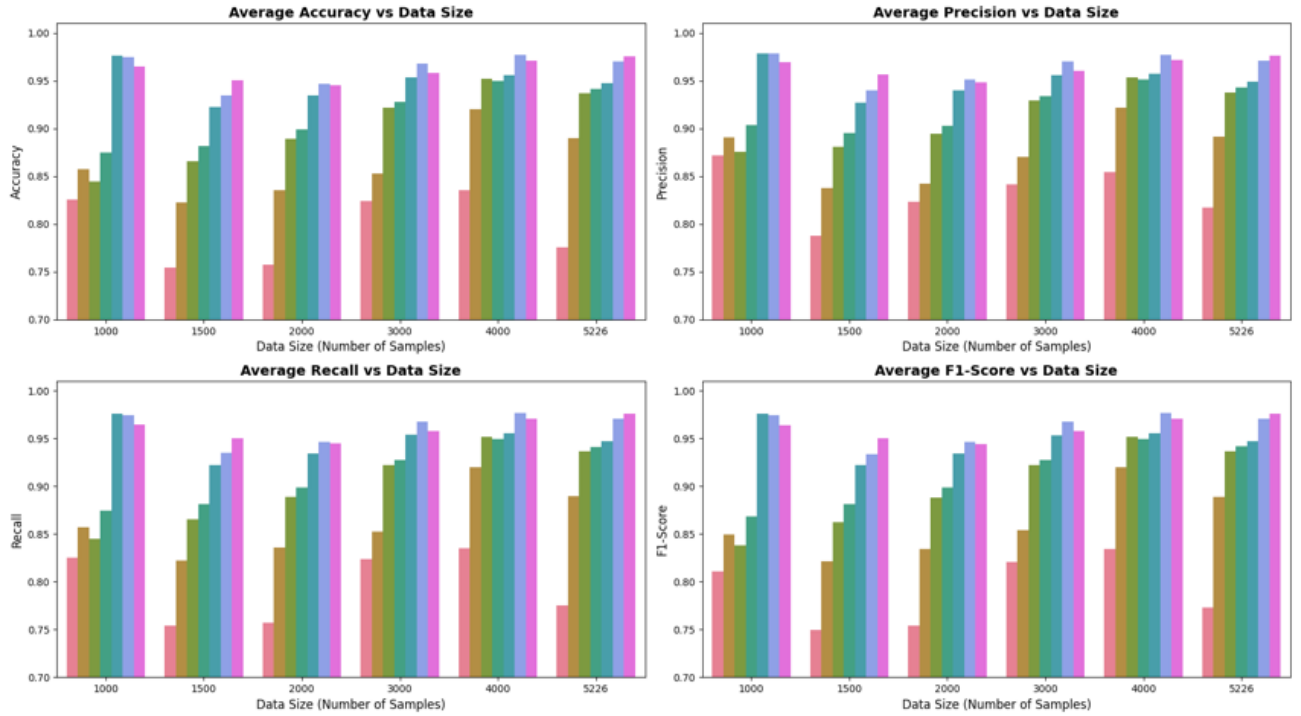


Figure 8: Performance comparison of ML models with varying training data sizes

features led to rapid and substantial gains in recognition accuracy, especially for ensemble and kernel-based models. The combination of gesture and identity features allowed models to better capture inter-user variations, leading to faster convergence and higher saturation points. Random Forest in particular scaled exceptionally well, achieving near-peak performance with moderate data increases. Linear models also showed improved data efficiency, reaching higher accuracy plateaus earlier than in the gesture-only experiments, underscoring the value of integrated feature representations for scalable and robust person-aware gesture recognition.

In gesture-only experiments (Experiments 1 and 2), Logistic Regression and SVM consistently achieved the highest accuracy, indicating that the extracted gesture features are largely linearly separable. Performance showed limited sensitivity to training-testing ratio variation, while dataset size had a more noticeable impact. Ensemble-based models, particularly Random Forest and XGBoost, benefited from larger datasets, whereas Naive Bayes consistently exhibited lower performance.

Incorporating person identification features (Experiments 3 and 4) resulted in a substantial improvement across all classifiers. Random Forest achieved the highest accuracy, reaching up to 99.26 percent, and demonstrated strong robustness and scalability. Logistic Regression remained highly stable and data-efficient, achieving high accuracy

even with limited training data. Most models reached performance saturation at relatively small dataset sizes, highlighting strong learning efficiency.

Overall, the results confirm that person-aware gesture modeling significantly enhances classification accuracy and reliability. Dataset size has a greater influence than training ratio, and ensemble models provide superior performance in complex, multi-class scenarios, while linear models offer an efficient and robust alternative for practical deployment.

6. Conclusion & Future Work

In this paper, we proposed and showed an efficient coordinate-based hand gesture recognition system using machine learning. Instead of using hand images for recognition, we applied hand landmark position for saving computational load. Based on the experiment results, spatial features from hand coordinate information made accurate and reliable gesture recognition system that suits for touchless application. However, issues such as variations of users, environments and dataset size remain. The future work will cover larger dataset, and use temporal modeling or deep learning for robust and scalable hand gesture recognition.

7. Declarations

The authors declare that there is no competing financial interests or personal relationships that could have affected the paper's integrity. The article involves no studies with human participants or animals conducted by any of the

authors. The authors have not received any disclosed funding for this research.

- **Conflict of interest:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Ethical approval:** This article does not contain any studies with human participants or animals performed by any of the authors.
- **Funding:** The authors have not disclosed any funding.
- **Data Availability:** The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to restrictions, e.g., their containing information that could compromise the privacy of research participants.
- **Ethical Considerations for AI Use :** AI tools were used in accordance with best practices, ensuring transparency, fairness, and avoidance of bias. No personally identifiable data were processed by AI systems without consent.
- **Informed Consent:** All participants provided informed consent prior to their inclusion in the study. Participation was voluntary, and confidentiality of all personal information was ensured.

8. Bibliography

CRedit authorship contribution statement

Sovan Bhattacharya: Conceptualization, Supervision, Data curation, Writing – original draft.. **Somnath Saha:** Data acquisition, Feature extraction, Dataset preparation, Writing – original draft.. **Dhritiranjana Nayek:** Real-time gesture tracking, Data preprocessing, Experimental evaluation, Writing – original draft.. **Basudev Saw:** Dataset validation, Duplicate handling, Comparative model analysis, Writing – original draft.. **Sumit Pal:** Methodology design, Model evaluation, Performance analysis, Writing – original draft..



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